

- 69** Coat colour variation in a particular population of mice is only affected by one gene with two alleles, R and r. This gene is not on a sex chromosome.

Heterozygous mice have yellow fur. Embryos that are homozygous dominant do not survive.

A yellow male and a yellow female mouse were mated several times and a large number of offspring were produced. Some of the offspring were grey in colour and others were yellow.

Assuming that no new mutations have occurred, which of the following is correct?

- A** 25% of the live offspring will be grey in colour.
- B** All grey mice have a homozygous genotype for coat colour.
- C** Offspring with XY chromosomes are all heterozygous for coat colour.
- D** The live offspring of a cross between a yellow and a grey mouse will always be yellow.
- E** There is a 3 : 1 ratio of dominant to recessive alleles for this gene in the live offspring.